

Quiz 4 CS221-2021-30 Minutes

Quiz 4 CS221

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Points: 33/40

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Name *

AADI AARYA CHANDRA

2

Roll Number

*

200101002

3

IITG email id

*

c.aadi@iitg.ac.in

✓ **Correct** 2/2 Points

4

For the given FSM in the state table (Figure of Question 4, 5, 6 and 7), in filling up the implication chart for the same, what will be values for question box 1

PS	NS		Output
	X=0	X=1	
A	D	C	0
B	F	H	0
C	E	D	1
D	A	E	0
E	C	A	1
F	F	B	1
G	B	H	0
H	C	G	1

	A	B	C	D	E
B					
C					
D			1 ?		
E			3 ?		
F		2 ?			
G					
H					4

Assume W-X
Y-Z

Implication Chart

- ☐ (E,D)/(A,E)
- ☐ (E,A)/(D,E)
- ☒ Crossed Sign ✓
- ☐ Non of the above

✓ **Correct** 2/2 Points

5

For the given FSM in the state table (Figure of Question 4), in filling up the implication chart for the same, what will be values for question box 2

- ☒ Crossed Sign ✓
- ☐ (F,H)/(F,B)

- ☐ (F,F)/(H,B)
- ☐ Non of the above

✗ **Incorrect** 0/2 Points

6

For the given FSM in the state table (Figure of Question 4), in filling up the implication chart for the same, what will be values for question box 3

- ☒ (E,A)/(C,D)
- ☐ (E,C)/(A,D) ✓
- ☐ Crossed
- ☐ (E,D)/(F,C)

✗ **Incorrect** 0/2 Points

7

For the given FSM in the state table (Figure of Question 4), in filling up the implication chart for the same, what will be values for question box 4

- ☒ Crossed
- ☐ (C,A)/(C,G)
- ☐ (C,A)/(F,H)
- ☐ (C,C)/(A,G) ✓

✓ **Correct** 1/1 Points

8

Which of the following is be the possible advantage of using one-hot encoding for FSM states compared to using the minimum number of bits:

- ☐ Number of flip flops will get reduced
- ☐ The state table would have lesser number of rows
- ☐ There are more parallel paths and hence the circuit is faster
- ☒ There is no need to decode the state for the output, therefore output combinational logic could be faster ✓

✓ **Correct** 3/3 Points

9

For the given FSM in the state table, determine the equivalence states using any methods

<i>PS</i>	<i>NS</i>		<i>Z</i>	
	<i>X=0</i>	<i>X=1</i>	<i>X=0</i>	<i>X=1</i>
<i>A</i>	<i>B</i>	<i>C</i>	<i>0</i>	<i>1</i>
<i>B</i>	<i>A</i>	<i>E</i>	<i>1</i>	<i>0</i>
<i>C</i>	<i>D</i>	<i>A</i>	<i>0</i>	<i>1</i>
<i>D</i>	<i>C</i>	<i>E</i>	<i>1</i>	<i>0</i>
<i>E</i>	<i>A</i>	<i>F</i>	<i>1</i>	<i>0</i>
<i>F</i>	<i>E</i>	<i>F</i>	<i>0</i>	<i>1</i>

- ☐ (A, B) and (C, D)
- ☒ (A, C) and (B, D) ✓
- ☐ (A, D) and (B, C)
- ☐ (A,C,D) and B

✓ **Correct** 2/2 Points

10

Given a FSM with 50 states. Implementation of the FSM requires x, y and z flip flops using binary encoding, one hot encoding and gray code encoding schemes respectively, then the value of $x + y + z$ will be ____.

✓ **Correct** 3/3 Points

11

For the given FSM in the state table, find which states are equivalent.

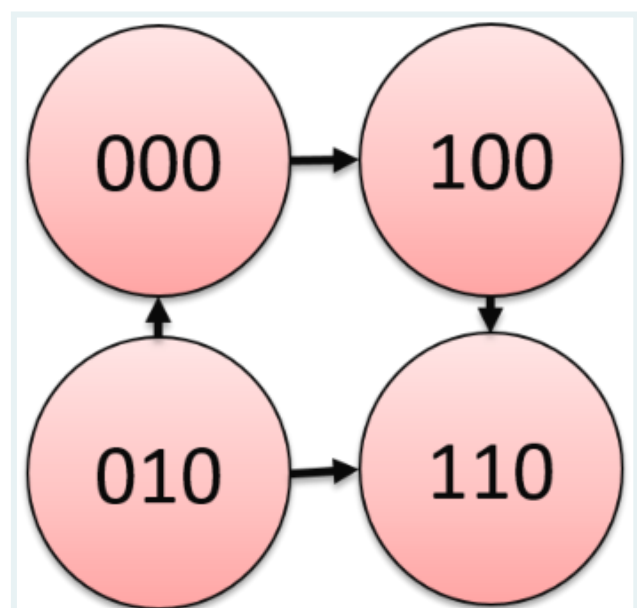
Present State	NS for X = 0,1	O/P for X = 0,1
a	b , d	0, 1
b	c , b	0, 1
c	a , d	1, 0
d	a , c	1, 0

- ☐ a,b
- ☒ c,d ✓
- ☐ a,c
- ☐ None of the above

✗ **Incorrect** 0/2 Points

12

You are allowed to choose the start-ups transitions that minimize the logic for the below state diagram using hamming distance minimization in designing 3 bit self-starting counter. What would be the minimum total hamming distance of such self-starting state diagram?



4

Correct answers:

8

✓ **Correct** 2/2 Points

13

1. Hamming Distance for two adjacent entries of state table of FSM in gray code encoding, one hot encoding and Johnson encoding respectively are?

☐ 1, 1 and 2

☒ 1, 2 and 1 ✓

☐ 2, 1 and 2

☐ 2, 2 and 1
✓ **Correct** 3/3 Points

14

After minimization the given FSM of the Table, the number of states remain is :

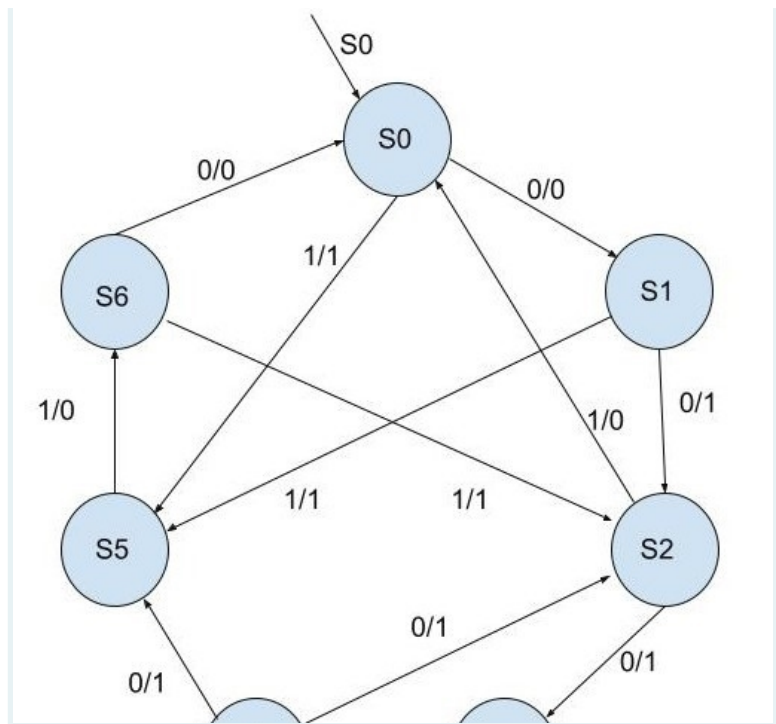
PS	NS		Output
	X=0	X=1	
A	D	C	0
B	F	H	0
C	E	D	1
D	A	E	0
E	C	A	1
F	F	B	1
G	B	H	0
H	C	G	1

6

✓ **Correct** 2/2 Points

15

Choose the correct option regarding the following FSM



- ☐ It is correct and complete
- ☐ It is not correct but complete
- ☐ It is correct but not complete
- ☒ It is not correct and not complete ✓

✓ **Correct** 1/- Points

16

How many flip-flops will be required in state register to implement a FSM with 15 states.

4

✓ **Correct** 1/1 Points

17

Choose the true option

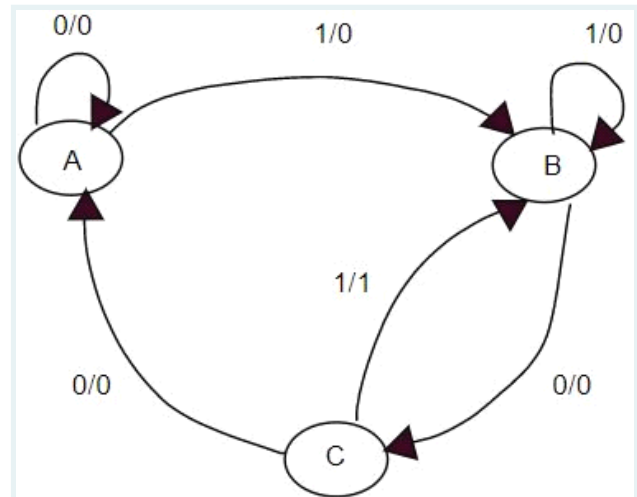
- ☐ Any given Mealy machine has an equivalent Moore machine

- ☐ Any given Moore machine has an equivalent Mealy machine
- ☐ Both Moore and Mealy machines are FSM's with output capability.
- ☒ All of the above ✓

✗ **Incorrect** 0/2 Points

18

A FSM with one input X and one output Z is defined by the state diagram below. Assume the FSM starts in state A. Given the following sequence of input X values. X : 1 0 0 1 0 1 1 0 1 0 1 0 (start from right to left). What state is the FSM in at the end of the above sequence? and What input sequence is detected by the FSM?



- ☒ State C, Sequence Detected : 010
- ☐ State C, Sequence Detected : 101
- ☐ State B, Sequence Detected : 101 ✓
- ☐ State B, Sequence Detected : 010

✓ **Correct** 2/2 Points

19

Are the states c and d equivalent?

Present State	NS for $x = 0$	O/P for $x = 0$

- ☐ Yes
- ☒ No ✓
- ☐ Maybe

✓ **Correct** 1/1 Points

20

Two state X and Y of a FSM are said to be equivalent if

- ☐ For every input sequence, the outputs are same but next states can be different
- ☐ For every input sequence, the next states are same but outputs can be different
- ☒ For every input sequence, the outputs are same and next states are same ✓
- ☐ For every input sequence, the outputs are well as next states can be different

✓ **Correct** 2/2 Points

21

What is the minimum number of states required to design a mealy machine which takes a 1-bit input and 2-bit output (Carry, Sum) such that output is the sum of the present and the previous bits of the input?

- ☐ 3
- ☐ 4
- ☒ 2 ✓
- ☐ 5

✓ **Correct** 2/2 Points

22

For the given FSM in the state table, find which states are equivalent

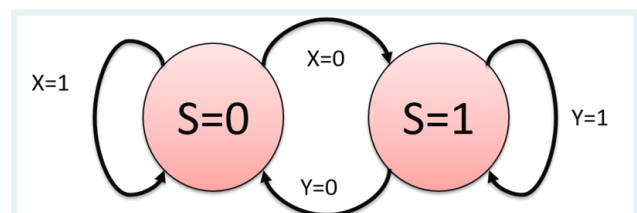
Present State	NS for X = 0,1	O/P for X = 0,1
a	b , c	1, 0
b	d , b	0, 0
c	a , d	1, 0
d	c , b	1, 0

- ☐ a,b
☐ c,d
☐ a,d
☒ non of the above ✓

✓ **Correct** 1/1 Points

23

For a state machine with the above state diagram the expression for the next state $S+$ in terms of the current state S and the input variable X and Y is

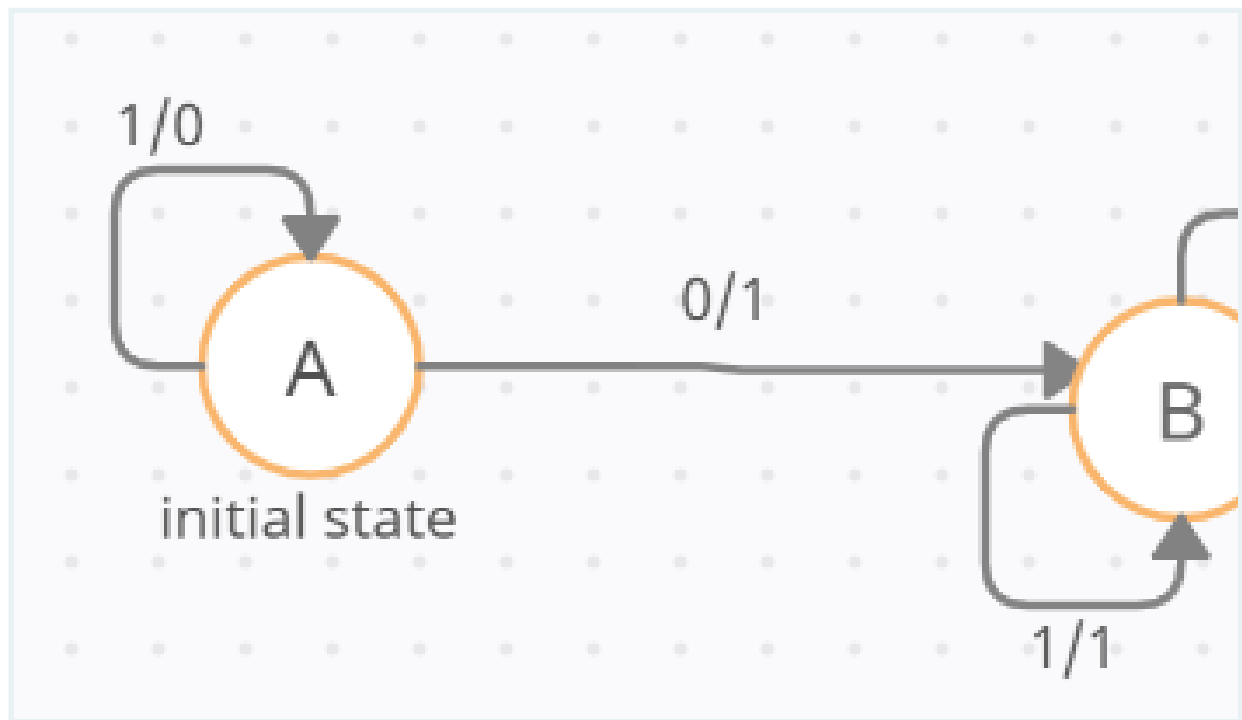


- ☐ $S+ = S'Y' + SX$ ✓
☒ $S+ = SXY' + S'YX'$
☐ $S+ = XY'$
☐ $S+ = S'Y + SX'$

✓ **Correct** 1/1 Points

24

Given the Mealy FSM, what is the output produced by the machine given a binary input 01011110? (Assume the binary input is read from the right side, similarly output read from right)



- ☐ 11011010
- ☐ 01011100
- ☒ 01011111 ✓
- ☐ 01011000

✓ **Correct** 2/2 Points

25

Suppose for a FSM, let the two states X and Y of the FSM, if two transitions from these two states have a common next state Z in the same FSM, then the best encoding for the states X, Y and Z among these following encodings will be:

- ☐ X(10000), Y(11001), Z(00000)
- ☐ X(10000), Y(11001), Z(11011)
- ☒ X(00000), Y(00001), Z(11011) ✓
- ☐ X(10000), Y(11111), Z(00000)

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